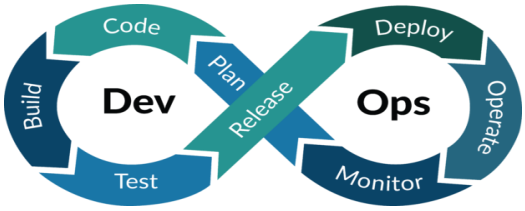

AWS DevOps Course Content

DevOps Introduction:

DevOps is **the combination of cultural philosophies, practices, and tools that increases an organization's ability to deliver applications and services at high velocity**: evolving and improving products at a faster pace than organizations using traditional software development and infrastructure management processes.

DevOps Syllabus:

1. **Linux**
   a) Introduction
   b) Working with basic commands
   c) Working with users and groups
   d) Working with file and directory
   e) Working with permissions
   f) Working with listing commands
   g) Working with change directory
   h) Working with utill commands
   i) Web service – apache httpd
   j) App Service – Apache Tomcat
2. **GIT**
   a) Introduction to Version Controlling
   b) GIT local and remote repositories
   c) Configuring git client

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d) Types of repositories – public and private
e) Working with repositories
f) Basic operations with git
g) Working with branches
h) Working with tags
i) Merge and rebase in git
j) Git log
k) Git revert
l) Token basis communication

3. Maven

a) Introduction to Maven
b) Why Maven Projects
c) Understanding dependency and dependent
d) Maven Installation
e) Types of maven repositories
f) Explanation on pom.xml file
g) Types of Applications
h) Maven Goals & Execution

4. Jfrog

a) Introduction to Jfrog
b) Installation of Jfrog
c) Understanding Artifactory Importance
d) Understanding Jfrog dashboard
e) Pushing Artifacts to Jfrog

5. Jenkins

a) Introduction to Jenkins
b) Installation of Jenkins
c) Plug-in management in Jenkins
d) User management in Jenkins
e) Permissions – Global, Item, Roles

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f) Jenkins Master Slave Configuration
g) Introduction to Groovy Scripting
h) Introduction to CI/CD
i) Writing pipelines for Deployment
j) Writing pipelines for configuration changes
k) Scripted pipelines
l) Declarative pipelines
m) Jenkins – Build Trigger types
n) Parallel Job Execution
o) Advantages of Multi Branch Pipelines (MBP)
p) Parameterized Jobs configurations
q) Scheduling builds automation
r) Integration Jenkins with other tools and services

6. Ansible – Configuration Management

a) Introduction to Ansible
b) Installation of Ansible
c) Configuration of Master and Nodes – s/d aws cli
d) Working with Static Inventory configuration
e) Working with Dynamic Inventory configuration
f) Working with host selection
g) Working with ansible.cfg file
h) Working with Ad-HOC commands
i) Working with Modules
j) Working with YAML Scripting
k) Working with playbooks – web servers, app servers
l) Working with handlers
m) Working with Roles & Galaxy
n) Working with – loops, conditions
o) *Project: GIT, Maven, Jenkins, Ansible, Web Servers*

7. Docker - Containerization

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a) Introduction
b) Difference between Docker and VM's
c) Uses of Docker
d) Installation of Docker Engine /Service
e) Working with Docker Files
f) Working with Docker Images
g) Working with Image tags
h) Pulling and Pushing Docker images to Docker Registry
i) AWS CLI communication for pushing Images
j) Pulling and Pushing Docker images to AWS ECR
k) Working with Docker Containers – platform & Service
l) Building Docker images on containers
m) Working with package management basics – yum, apt, pkg, dnf
n) Working with Docker Volumes
o) Working with Docker Networking
p) Working with container communication
q) Working with Docker Compose
r) Working with Docker Swarm
s) Project: Integration with Jenkins & implementation as project

8. Kubernetes

a) Introduction
b) Understanding about Kubernetes Architecture
c) Kubernetes Environment setup
d) Understanding Cluster, Node group, Node, Pod, Service
e) Writing deployment files using YAML
f) Writing Service files using YAML
g) Understand about restart and scaling policies
h) Describe, edit, login into pods
i) Project: GIT, Maven, Jfrog, Jenkins, Ansible and Kubernetes

9. Terraform

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a) Introduction to Terraform
b) Terraform providers
c) Terraform setup
d) Working with AWS Provider
e) Working with creating EC2 resource
f) Working with creating VPC resource
g) Working with creating Subnets resource
h) Working with creating Route Table resource
i) Working with creating Internet Gateway resource
j) Working with creating Security Groups
k) Working with creating key pairs
l) Terraform state files and backup state files
m) Terraform Variables
n) Project: GIT, Jenkins, Terraform

10.Helm Charts

a) Introduction to Helm charts
b) Architecture
c) Installations
d) Commands
e) Charts
f) Templetes
g) Creating Custom Helm Charts
h) Demo: Installing Bitnami wordpress Helm chart

11.ELK

a) ELK Introduction
b) Installation
c) Oberserbility

12. Datadog

a) DataDog Introduction
b) Installation

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c) Kubernetes Monitoring with the Datadog

13.Prometheus And Grafana

a) Introduction – Prometheus
b) Monitoring
c) Alerting
d) Introduction – Grafana
e) Installation
f) Data sources in Grafana
g) Dashboards & Explore in Grafana

14.AWS – Amazon Web Services

- Cloud Introduction – Deployment models
- IAM – Identity and Access Management
- AWS CLI – Command Line Interface
- EC2 – Elastic Compute Cloud
- Lightsail
- Elastic Beanstalk
- Auto Scaling
- ELB - Elastic Load Balancer
- AWS Lambda Service
- ECR – Elastic Container Registry
- ECS – Elastic Container Service
- EKS – Elastic Kubernetes Service
- S3 – Simple Storage Service
- RDS – Relational Database Service
- EBS – Elastic Block Storage
- VPC – Virtual Private Cloud
- AWS Cloud front
- AWS Amplify
- SNS – Simple Notification Service

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- Route 53
- Billing info

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